

The “Architectural Jewels of Lublin” Game as a Tool for Collaborative Interactive Learning of History

Marek Milosz^(✉) and Jerzy Montusiewicz

Lublin University of Technology, Lublin, Poland
{m.milosz, j.montusiewicz}@pollub.pl

Abstract. The city of Lublin in Poland is a place of many overlapping cultures and religions: Polish, Russian, Ukrainian, Armenian, Jewish, German, etc. Remains of these cultures are present in the city space in the form of architectural monuments. Unfortunately, the awareness of this fact disappears in society, especially among the school youth.

The paper presents the results of a survey of the level of knowledge about the architectural monuments of the city of Lublin, Poland. The study was conducted among K14 youngsters with the use of questionnaires and 3D models. The results of the research show a low level of knowledge about monuments and their history.

The next part of the article presents the game “Architectural Jewels of Lublin”, which was developed as a board game, using modern information technologies such as RFID systems, 3D modelling and printing, gamification and real time systems programming. The technical layer of the game is an electronic board with the city plan of Lublin and models of architectural monuments, a sensory network using RFID technology, and a palmtop with game control software. The game allows the implementation of collaborative interactive learning of history. The article also presents the logic of the game and the elements of gamification used in it. The directions of its development and dissemination are also presented.

Keywords: Interactive learning · Board game · Knowledge level
Architectural monuments

1 Introduction

The city of Lublin is the largest Polish city on the eastern side of the Vistula River. Slavic settlement in Lublin is dates back to the end of the 8th century. The favourable geographical location on the numerous hills above the Bystrzyca and Czechówka valleys, at the crossroads of trade routes, allowed for a rapid development of the castle in the following centuries and the arrival of representatives of many different nationalities. In 1317, in order to reorganise the existing state of affairs and strengthen further development, the city was re-located on the Magdeburg law, well-known and commonly used in Western Europe.

At present the inhabitants of Lublin celebrate the 700th anniversary of this very important event. Over the centuries, Lublin and the whole Polish state were an area in which Poles, Ruthenians, Russians, Jews, Armenians, Germans, Ukrainians and other

ethnicities lived side by side in symbiosis and the spirit of tolerance. Alongside each other, there were opportunities for different cultures and religions to develop: Catholic, Orthodox, Uniate (Greek-Catholic), Protestant and Jewish. This situation persisted until Poland’s loss of independence in the 18th century, when the Uniates were persecuted by the Russian authorities. During the Second World War, the German occupation authorities exterminated the Jewish population, destroying many of its historical objects.

Multinationality, multiculturalism and multiconfessionality are the standard image of eastern Poland, as exemplified by Lublin. Remnants from those centuries are numerous preserved architectural monuments, which testify to both the specific character of the merchant town and the wealth of its inhabitants, and the foundations made by the Polish kings and surrounding nobles.

Unfortunately, as experience shows, young people (especially in the younger age bracket, K14), have little historical awareness of Lublin’s rich history. This is reflected in the poor recognition of monuments (including architectural ones) that remain after the people once living in the city.

In view of the fact that in 2017 the city of Lublin celebrates 700 years of its location, in the Institute of Computer Science of the Lublin University of Technology arose the idea of building a board game entitled “Architectural Jewels of Lublin”, which in a modern way would allow to raise the level of pupils’ knowledge about the architectural monuments of Lublin [1]. The developed game realises the idea of gamification, which assumes that in addition to the competition phase, winning points and prizes by participants. It also allows the transmission of strictly defined content. In preparation of the game, advanced 3D modelling and printing technology was used, as well as automatic recognition of the 3D model’s location on the map of the old city. The whole process of competition is managed by real-time systems programming.

Games are a very popular educational tool used in different areas and levels of education [2–12]. The effectiveness of their use is confirmed by numerous scientific studies [13, 14] and the opinions of participants, which indicate [15, 16] their interest and motivation, the intensification of learning, the approximation to reality, the inter-disciplinarity, the development of practical skills without the usual risks associated with such activity, interaction and error learning.

Boards have been used to organise games for thousands of years, examples being as chess, sudoku, or war games. Also, attempts to automate board games have been made since ancient times. The development of board game automation technology led from mechanical, through electromechanical to electronic one, using a variety of component bases and the use of various physical phenomena to build sensory networks [17–19]. Contemporary computer technologies provide a wide range of opportunities for automated board games [20–22], which was used in the “Architectural Jewels of Lublin” game [23].

The article consists essentially of two main parts. Section 2 presents the research on the level of knowledge about architectural monuments, conducted among young people of K14 from the city of Lublin and the region. The research was carried out in a classical format: defining goals, formalising them, conducting surveys, processing results, and discussing them. Section 3 presents the game “Architectural Jewels of Lublin” in terms of: game

objectives, technical layer and game execution logic. The article ends with conclusions from the achievements so far and indicates further work.

2 Study of the Level of Knowledge About the Architectural Monuments of the City of Lublin Among Young People

2.1 Purpose and Scope of Research

The aim of the study was to assess the level of knowledge of K14 pupils of the architectural monuments of the city of Lublin. The study covered two groups of pupils: those living in Lublin (1st group) and those outside (2nd group).

The level of knowledge was assessed on the basis of the results of the test, consisting of questions about the 7 most distinctive architectural monuments of the city of Lublin, selected by experts. For each monument the questions concerned:

- P1. Name of the monument.
- P2. The time at which the monument originated.
- P3. The architectural style in which the monument was built.

For the level of knowledge about architectural monuments of the city of Lublin the following absolute indicator (K) was adopted:

$$K = \sum_{i=1}^7 \sum_{j=1}^3 w_j P_{ij} \quad (1)$$

where:

- i – number of the monument,
- j – number of the question for the given monument,
- w_j – the weight of the j -th question for the monument,
- P_{ij} – the answer to the j -th question for the i -th monument (0 - incorrect, or 1 - correct).

The following weights for individual questions (P1... P3) were used for each monument: P1 – 1, P2–2 and P3–3. These scales (1... 3) in the opinion of experts mirror the scale of the difficulty of the questions.

The relative level of knowledge (k) is the ratio of a student's absolute knowledge index to its maximum value – K_{max} :

$$k = \frac{K}{K_{max}}, \% \quad (2)$$

For 7 monuments and 3 questions of different difficulty levels, $K_{max} = 42$.

2.2 Thesis and Research Hypotheses

The analysis of the research objective allowed us to formulate the following research thesis:

Pupils from the K14 group living in the Lublin province have little knowledge about the architectural monuments of the city of Lublin.

In order to confirm this thesis, two hypotheses were formulated:

H1. The average level of knowledge of K14 pupils living in the Lublin province about the architectural monuments of Lublin is very low.

H2. The level of knowledge of K14 pupils living in the Lublin province about the architectural monuments of Lublin practically does not depend on whether the student lives in or outside Lublin.

2.3 Research Methodology

The research group consisted of pupils who participated in a visit to the Lublin University of Technology. All participants were aged K14.

In order to carry out the research, a survey was prepared, consisting of a questionnaire (containing answers classifying the respondent) and questions on the architectural monuments of Lublin. For each of the seven monuments, 3 questions were asked in Sect. 2.1.

The study was conducted on two groups of pupils living in the Lublin province. The pupils from the first group of respondents were permanent residents of Lublin from birth, while those from the other group were not. The size of the two groups was the same: 12 pupils. Survey participants were randomly selected – each from a different school in Lublin and the region.

The research was carried out using paper surveys and printed 3D models of architectural monuments on a scale of 1:200. After showing a model selected from the questionnaires of all models, the pupils answered 3 questions (P1... P3) about the model shown. After the last question the polls were collected, and the researcher gave correct answers about each of the monuments, along with its extended historical background.



Fig. 1. Implementation of the research on pupils' knowledge about the architectural monuments of Lublin

This presentation was a reward for the pupils and was an element of building historical knowledge (Fig. 1).

2.4 Results

The results of the questionnaires were transformed according to Formulas (1) and (2) – for each student a relative level of knowledge was determined. In Table 1 the basic statistics of the survey results are presented.

Table 1. Results of questionnaires on the statistical parameters of the relative level of respondents’ knowledge about architectural monuments of Lublin

No.	Name	Value <i>k</i> for all pupils	Value <i>k</i> for pupils from outside Lublin
1	Average	23.1%	20.2%
2	Median	26.2%	20.2%
3	Min.	0.0%	0.0%
4	Max.	50.0%	40.5%
5	Standard deviation	13.2%	13.7%

A more accurate statistical analysis of the results of the study was made by defining the series of distributions and constructing histograms of the distribution of the values of the relative level of knowledge of pupils about the architectural monuments of Lublin. Its results are presented in Fig. 2.

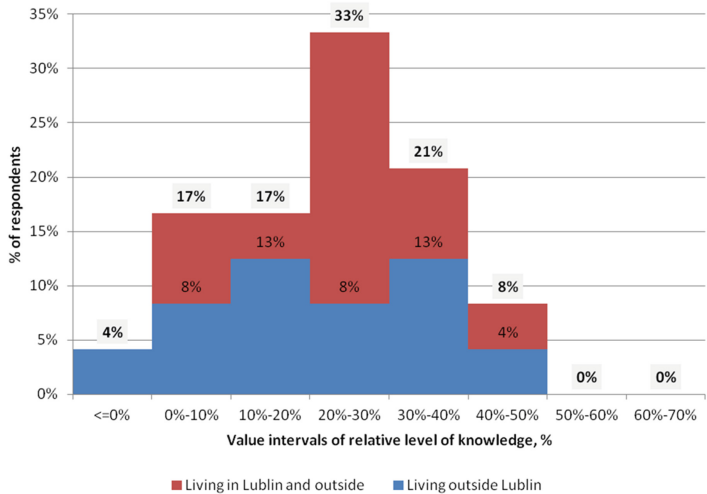


Fig. 2. Histogram of distribution of the relative level of knowledge of pupils living in Lublin and those from outside it

2.5 Discussion

The very low mean (and median) values of the relative knowledge level of pupils, attaining a little over 20% (Table 1), clearly indicate a low level of pupils’ knowledge of Lublin’s architectural monuments. This is also confirmed by the fact that no student exceeded the 50% threshold, usually required to pass the course.

Hypothesis H1 has been positively verified

The mean of the pupils’ self-esteem of their level of knowledge about the monuments of Lublin amounted to 1.96 in the scale of 1... 4 (where 1 – very bad, 4 – very good), whereas for the pupils from outside Lublin this value was 1.41, and for those living in Lublin – 2.50. Nevertheless, the comparison of the average level of knowledge of all pupils with the average level of knowledge of non-Lublin pupils (Table 1 and Fig. 2) indicates that these levels are not much different from one another.

Therefore hypothesis H2 can be considered as positively verified

The results of the K14 pupils’ performance showed a low level of knowledge about Lublin’s architectural monuments and their history.

3 The Game “Architectural Jewels of Lublin” – Presentation

3.1 Idea of the Game

The game “Architectural Jewels of Lublin” is to provide participants with the most distinctive and famous monuments of Lublin, their history and location in urban space. In its basic form, the game is designed for two players at the same time. Players aim to score as many points as possible and thus defeat a direct opponent. Points can be earned for proper positioning of the object on the board and correct answers to questions. Competitors have the following interactive features:

- 3D models of Lublin monuments,
- the “I answer” button is used to take over question 2,
- buttons to answers (1/2/3).

In addition, there is a palmtop on the board that displays questions, possible answers, historical information about the object being held, current game results and the final score, as well as the loudspeakers giving the relevant messages.

The large board and 3D models, combined with a palmtop with control software, facilitate the collaborative and interactive capabilities of the participants in the game.

3.2 Technical Layer

The game board is a prepared and printed 125 × 142 cm plan of the old city, placed on a specially designed support structure. The visible part of the board contains a simplified layout of streets and squares of the historical centre of the city with places where 3D models of architectural monuments in Lublin are marked. At these locations, a LED

RGB diode will light up in green when the model is placed correctly, or in red in the case of a mistake, Fig. 3. The board also has a palmtop responsible for managing the entire game, retrieving information about the current state of layout of the models on the board. It displays current attendee messages, questions and historical information, and a set of loudspeakers to generate sound messages.

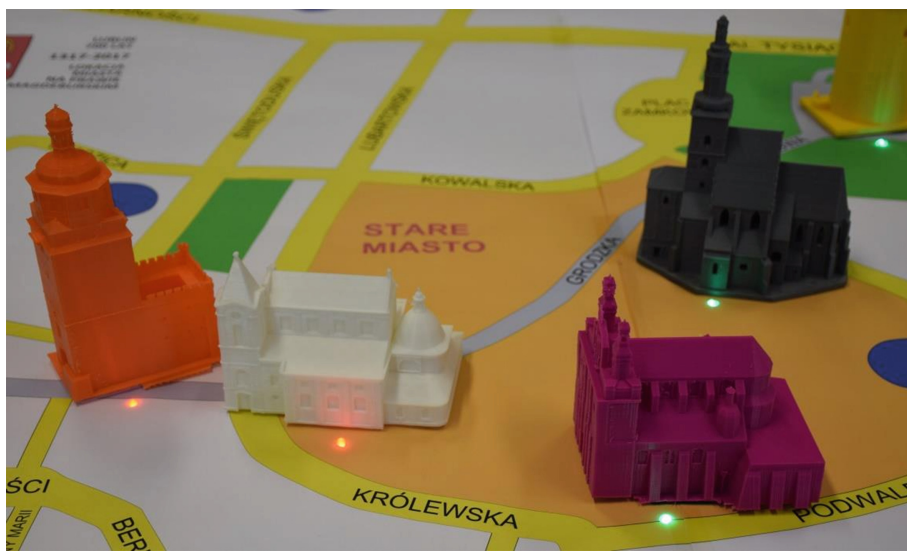


Fig. 3. The game board with examples of 3D models of Lublin's architectural monuments placed as a demonstration

Invisible parts of the board are RFID readers, whose task is to identify the correctness of 3D model placement, Arduino boards responsible for controlling readers and RGB LEDs, and for transferring their current state to the software controlling the whole game, wiring (over 130 m), power strip and power supplies.

3.3 The Logic of the Game Allowing for Collaborative-Interactive Learning of History

The game is turn-based and consists of repeated cycles sequentially played by players. Each cycle includes:

- retrieving a 3D model from the bank,
- placing it on the board,
- answering question 1 (concerning the object),
- answering question 2 (about the city's history). This question can be taken over by the opponent.

The gameplay and its course are dynamic, controlled by the real-time software developed. Depending on the situation, the messages displayed on the screen change

and the course of the current turn is appropriately modified. In the course of the game different situations are possible (Table 2).

Table 2. The scheme of the course of a single turn of the game

No.	Occurrence	Action	Consequences
1	Correct placement of the model	The turn is continued	Display of question 1
2	Incorrect placement of the model or exhausting the time limit	Placement of a model into the object bank	No question 1. The opponent starts a new cycle
3	Wrong answer to question 1	Starting a new cycle	The opponent starts
4	No answer to question 1 – time out	The answer is given by the opponent	The opponent starts a new cycle
5	Wrong answer to question 2	Starting a new cycle	The opponent starts
6	No answer to question 2 – time out	The answer is given by the opponent	The opponent starts a new cycle

The number of points scored by a player depends not only on models correctly placed on the board and correct answers, but largely on the number of mistakes made by players. When a model is placed incorrectly and must return to the object bank, it becomes “more valuable” – it gets more points than it had. It is assumed that it is more difficult to recognise. The opponent starting the next cycle of the game can take a chance and choose it, or download another model with an initial score. The number of potential places for placing models on the board is much higher than the number of available models. This is to prevent the last model from being placed correctly merely by default.

4 Conclusions and Further Work

Survey research on a selected research sample has shown that K14 youths have a very low knowledge of the history of the city of Lublin, and in particular of the historical monuments built in the past. To improve this situation, the computerised board game “Architectural Jewels of Lublin” was developed. The large size of its board and the need to manually handle large 3D architectural monuments by many participants makes this game interactive and collaborative. The rules of gamification were used to create algorithms for individual turns and the whole game.

Further work related to the “Architectural Jewels of Lublin” game will focus on:

- Development of software for game control, mainly for elements that increase the dynamics of the turn and increase the educational value of the game.
- Study of the effectiveness of learning by using games – this efficiency, understood as the increase of knowledge in the participants of the game, can be analysed by conducting an active experiment with the participants (experiment: survey-game-play-survey).
- Increasing game difficulty by introducing models of non-existent monuments.

- Improving the realism of 3D models of monuments by applying realistic colours to their surfaces.
- Examination of the ergonomics of the game control interface software.
- Development and implementation of adaptive algorithms, modifying the way points are calculated depending on the course of the game.
- Increasing the number of players from the current 2 to 4, which will allow to develop completely different rules of the game, increasing its collaborative value on the basis of playing in pairs (similar e.g. to bridge).

The technical and programming solutions developed in the creation of the game “Architectural Jewels of Lublin” can be easily used to build analogous games related to the geolocation of monuments, cities, rivers, national parks, etc. or historical events (e.g. battles). The content layer of such games is of course unique, but the solutions developed on this occasion can easily be used in new circumstances.

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